

PROJECT ZENITH — CELESTIAL EYE README

The Celestial Eye — Real-Time Space Intelligence Dashboard

Project Zenith is a cinematic, real-time cosmic radar that tracks satellites, celestial bodies, and space events — built with custom WebGL shaders, React Three Fiber, and live space APIs.

■■ Installation and Setup Instructions

Clearly explained instructions to clone, install dependencies, and run the project locally on your machine.

Prerequisites: To build and run this application locally, you need the following utilities installed:

Requirement	Version	Link
Node.js	18.x or higher	https://nodejs.org
npm	9.x or higher	(Bundled with Node.js)
Active Internet Connection	Required	(For CDNs, geocoding & live telemetry)

Step-by-Step Local Deployment

1. Clone the project repository. Open your terminal and clone the source code:

```
git clone https://github.com/harshith7002/Project-Zenith.git
cd Project-Zenith
```

2. Install runtime and development dependencies:

```
npm install
```

3. Run the development server:

```
npm run dev
```

4. View the application:

Open your browser and navigate to **http://localhost:3000**. The page will auto-reload when you save changes.

Production Compiling & Optimization

To test the application under local production conditions (runs ESLint and type safety checks):

```
npm run build
npm start
```

■ Website Functionality and Unique Features

Project Zenith is built with custom WebGL, canvas rendering, and real-time computations to ensure the UI feels alive, responsive, and astronomically accurate.

1. Cinematic 3D Earth (WebGL & Custom Shaders)

- **Day/Night Terminator:** A custom fragment shader blends day and night textures dynamically based on the Sun's coordinate vector using smoothstep(-0.25, 0.25, cosTheta), creating a realistic glowing sunset twilight zone.
- **Golden City Lights:** Rendered on the night hemisphere of the globe and masked dynamically by cloud layers.
- **Ocean Specular Reflections:** Real-time specular sun-reflection reflections computed using per-pixel specular mapping.

- **Razor-Thin Atmospheric Corona:** A custom BackSide atmospheric glow shader renders a thin, bright blue rim on the sunlit limb, fading smoothly into the darkness of space at the terminator.
- **Organically Twinkling Starfield:** 1,000 circular stars rendered across two depth layers using a custom vertex-shader phase formula to slowly twinkle a subset of stars, eliminating visual noise.
- **Anamorphic Lens Flare:** Ray-occlusion tested so that the solar flare fades out naturally when the Sun is eclipsed by the Earth's geometry.
- **Drift & Mouse Parallax:** Slower, high-inertia camera parallax linked to mouse coordinates to create a dramatic, cinematic IMAX camera feel.

2. Geocoded Search & Autocomplete Navigation

- **Autocomplete Search Bar:** Allows users to search for any city or country. Queries OpenStreetMap's Nominatim API in the background to suggest matching locations dynamically.
- **Raw Coordinate Input:** Directly parses coordinates (e.g. 17.40, 78.37) typed into the search bar, immediately triggering updates and flying the globe.
- **Cache-First GPS Option:** Keeps geolocation as a secondary button beside search, caching coordinates in local storage and querying with maximumAge Infinity for <10ms response locks.

3. Celestial Radar & Mission Control Dashboard

- **ISS Tracker:** Fetches altitude, velocity, and coordinates of the ISS directly from unauthenticated REST endpoints, updating every 5 seconds.
- **Calculated Sky Quality Score:** Evaluates live cloud cover, humidity (from Open-Meteo API), Moon phase, and Bortle index. Utilizes pulsing skeleton loading during coordinate transitions.
- **Visible Sky Objects:** Performs Alt/Az horizon filters using astronomy-engine. Only displays objects physically above the horizon.
- **Pulsing Satellite Radar:** Projects orbits of Starlink, GPS, and weather constellations onto a 2D polar projection SVG radar grid.
- **Context-Aware AI Space Guide:** Explains night-sky details and generates natural language summary greet texts based on current coordinates.
- **Cosmic Event Predictor:** Live timers relative to the active coordinate meridian tracking eclipses, alignments, and meteor shower peaks.

4. Interactive 3D Globe

- Plots the live tracking path of the ISS in real-time.
- Clicking any coordinate dispatches coordinates instantly to update the dashboard and dispatches camera movements without requiring manual confirmations.

5. Solar System Simulator

- Renders all 8 planets orbiting the Sun in real-time 3D with Cassini Division Saturn rings and educational planet click modals.

■ Dependencies

External libraries, frameworks, and tools used to build Project Zenith:

- **Next.js** (15.5.19): React framework driving the SSR (Server-Side Rendering), App Router routing, and bundler.
- **React & React DOM** (19.1.0): Declarative component state and rendering structure.
- **three** (^0.184.0): Core WebGL/3D library for geometries, Custom ShaderMaterial compilations, lighting, and textures.

- **@react-three/fiber** (^9.6.1): React wrapper for rendering Three.js scenes declaratively.
- **@react-three/drei** (^10.7.7): Helper hooks and components for Three.js (controls, textures, points).
- **@react-three/postprocessing** (^3.0.4): Post-processing pipeline enabling glowing WebGL Bloom overlays.
- **react-globe.gl** (^2.38.0): Specialized wrapper for compiling the interactive coordinate mapping globe.
- **astronomy-engine** (^2.1.19): Calculates real-time planetary orbits, Alt/Az coordinates, and moon illumination phases.
- **satellite.js** (^7.0.1): Implements SGP4 orbit propagation to calculate satellite coordinates from Two-Line Element (TLE) datasets.
- **framer-motion** (^12.40.0): Declarative animation engine driving transitions and fades.
- **lenis & gsap**: Smooth scrolling integrations and timeline animations.
- **tailwindcss** (^4.x): Utility styling framework.